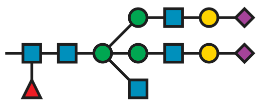
Amino acids

PEPTIDE PEPTJDE PE(?PTI)DE	Amino acids are written as the standard IUPAC symbols. With O/U added as pyrolysine, and selenocysteine. J/B/Z are ambiguous amino acids for I/L, N/D, and Q/E respectively. Lastly X is added as any unknown amino acid. X is defined to have a mass of zero. So 'X[+435]' can be used to indicate a mass gap of 435 Da.

Modifications

Modifications are defined between square brackets '[mod]'. Multiple definitions can be combined with a vertical line '|' like so '[mod|mod|mod]'.

Monoisotopic mass	PEPTI[+15.995]DE
Molecular formula	PEPTI[Formula:O]DE
Name (Unimod/PSI-MOD)	PEPTI[Oxidation]DE
Name specific ontology	PEPTI[U:Oxidation]DE
Index specific ontology	PEPTI[UNIMOD:35]DE
Comment or description	PEPTI[INFO:text]DE
Glycan composition	PEPN[Glycan:HexNAc5Hex5dHex1NeuAc2]ITDE
Glycan structure	PEPN[GNO:G75079FY]ITDE → Encodes:
Charged molecular formula	PEPTI[Formula:H-2Zn1:z+2]DE



Modification prefixes

Mass	Name	Index	Description
U	U?	Unimod	Modifications from Unimod
M	M?	PSIMOD	Modifications from PSI MOD
R	R	Resid	Modifications from Resid
X	X	XLMOD	Modifications from XL-MOD
G	G	GNO	Modifications from GNOme
Obs	-	-	To indicate an observed mass
-	INFO	-	To add comments or description
-	Glycan	-	Glycan compositions
-	Formula	-	Molecular formulas

Locations

Localised	
PEPT[mod][mod]IDE	(Multiple) modifications on the side chain of an amino acid
[mod][mod]-PEPTIDE	(Multiple) modifications on the N terminus
PEPTIDE-[mod][mod]	(Multiple) modifications on the C terminus
{mod}PEPTIDE	Labile modification, on this peptidoform
<mod@E,N-term>PEPTIDE	Fixed modification
Unlocalised	
[mod]^n?PEPTIDE	On the entire peptidoform
PEPT[mod#name]ID[#name]E	On the given locations
PEP(TID)[mod]E	Within the given stretch
Unlocalised additional rules	
Position:E,N-term	Limit the allowed positions, uses the same positions as fixed modifications
Limit:2	Limit the maximal number on one site, only for ^n global unlocalised modifications
CoMKP	Allows colocalising with placed modifications
CoMUP	Allows colocalising with other unlocalised modifications

The order of all options when used together

<mod@E>[mod]^n?{mod}[mod]-PEP-[mod]

Using unlocalised rules

PEP(TID[mod])[mod|Position:T,D|CoMKP]E

Combining peptidoforms

PEPTIDE+PEPTIDE	Chimeric peptidoforms (forming one compound peptidoform ion)
PEPT[mod#XLname]IDE//PEPT[#XLname]IDE	Cross-linked peptidoforms (forming one peptidoform ion)
PEPT[mod#BRANCH]IDE//PEP[#BRANCH]TIDE	Branched peptidoforms (forming one peptidoform ion)
PEPT[mod#XLname]IDE[#XLname]	Intra linked cross-linked peptidoform

Charge

PEPTIDE/2	Global charge, assumed to be protons
PEPTIDE/[Na:z+1,H:z+1]	Defined charge carriers, in this case 1 unlocalised sodium and 1 unlocalised proton
PEP[Formula:Na:z+1]TIDE/1	Defined charge carriers, in this case 1 localised sodium and 1 unlocalised proton, this still has a global charge of 2

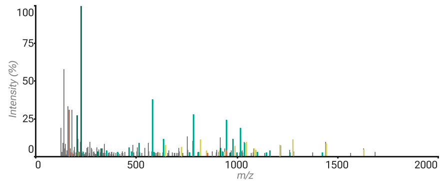
Name

To add a description or metadata to an entire (compound) peptidoform (ion). To do this for single locations or stretches use the INFO tag.

(>Peptidoform name)PEPTIDE
(>>Peptidoform ion name)PEPT[mod#XLname]IDE//PEPT[#XLname]IDE
(>>>Compound peptidoform ion name)PEPTIDE+PEPTIDE

Universal Spectrum Identifier

mzspec:<ID>:<file>:scan:<number>:<peptidoform>
index:<number>
nativeId:<number>(,<number>)*



mzspec:PXD023419:Peng2021_Herceptin_aLP.raw:scan:11368:RWGGDGFYAM[Oxidation]DYWGQGTLVTV/3

USIs are the URL for the mass spectrometry world. A USI points to a Proteomics Exchange Identifier (PXD...), a file, and a spectrum in that file. Additionally, a peptidoform can be added to the end, note that many servers require the peptidoform and a global charge state for the peptidoform.

ProForma v2.1
psidev.info/proforma
USI
psidev.info/usi

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V1.0